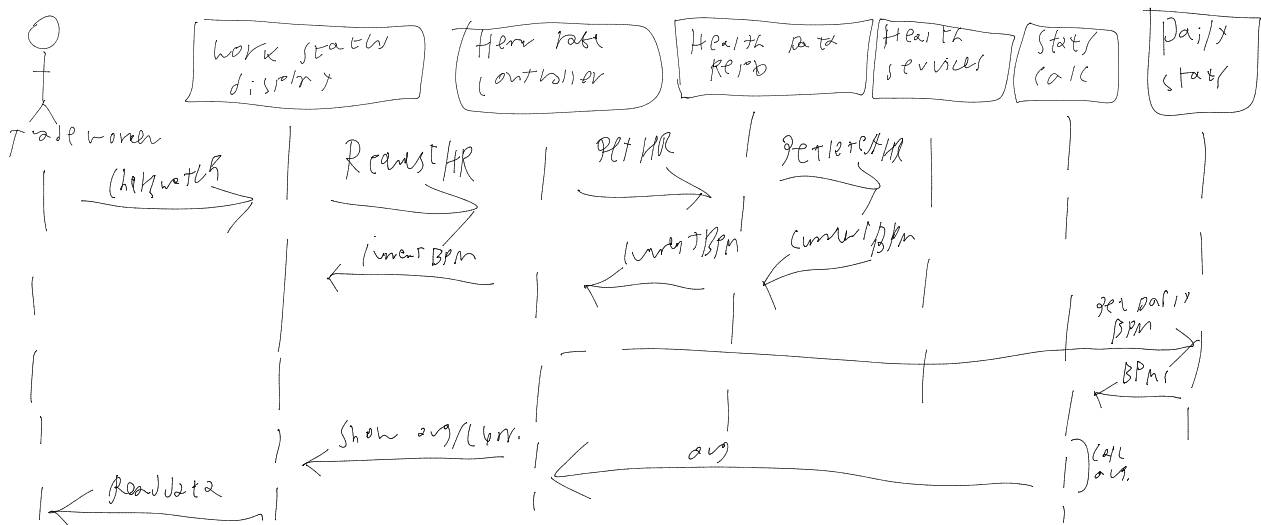


Diagram 1: Trade Worker BPM Display



avg and (current BPM Displayed

Visual indicator for significant deviation
Time from last reading

INCLUDE VARIABLES AND ARGUMENTS



Purpose

Models the interaction flow when a trade worker glances at their smartwatch to view their current heart rate and daily average BPM.

Preconditions

1. Trade worker has WearOS watch powered on and worn correctly
2. HeartRate monitoring app is running and has been active since start of workday
3. HealthServices permissions granted (BODY_SENSORS)
4. At least 5 BPM readings collected during current workday
5. Current time is during work hours (e.g., 10:34 AM)

Postconditions

What the Actor Sees:

1. **Primary display:** Current BPM "125" in large, bold font (32sp)
2. **Secondary display:** "Daily Avg: 98 BPM" in smaller font (18sp)
3. **Status indicator:** Yellow/orange color indicating ELEVATED status
4. **Timestamp:** "Updated 10:34 AM" in small gray text
5. **Visual feedback:** Subtle haptic feedback confirming display update

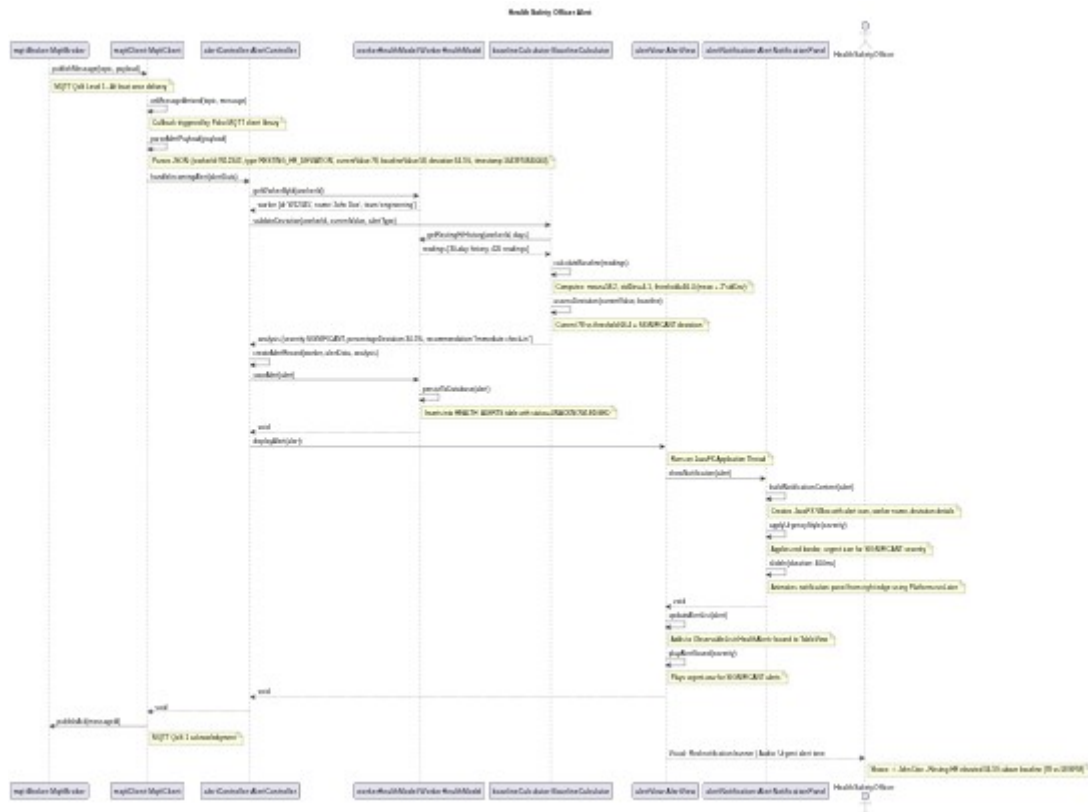
Data State Changes:

1. healthDataRepo internal cache updated with latest HeartRateReading object
2. No persistent storage writes (read-only operation for display)
3. workStatusDisplay.currentModel updated with new BpmDisplayModel
4. dailyStats object maintains in-memory collection (no new data added)
5. View rendering state refreshed in WearOS display buffer

System State:

- Background HealthServices monitoring continues
- Next automatic refresh scheduled in 60 seconds
- Battery-efficient passive monitoring remains active
- App lifecycle in RESUMED state

Diagram 2: Health & Safety Officer Alert



Purpose

Models the interaction flow when the Manager Portal (JavaFX application) receives and displays a health alert from a worker's smartwatch via MQTT, notifying the Health & Safety Officer of a significant biometric deviation.

Preconditions

1. Health & Safety Officer is logged into Manager Portal (JavaFX application)
2. Manager Portal is connected to MQTT broker at `mqtt://alerts.company.com:1883`
3. Officer is subscribed to topic `health/alerts/team/{teamId}`
4. Worker baseline data exists in system (minimum 30 days of resting HR data)
5. Worker's smartwatch has detected anomalous resting heart rate and published alert
6. Officer has Manager Portal window in focus or running in background

Postconditions

What the Actor Sees:

1. **Primary Alert Banner** (top-right of screen):
 - Red urgent border with warning icon ⚠️
 - Worker name: "John Doe"
 - Alert message: "Resting HR elevated 34.5% above baseline"
 - Metrics: "Current: 78 BPM | Baseline: 58 BPM"
 - Timestamp: "2 minutes ago"

- Two buttons: [Acknowledge] [View Details]

2. **Alert Table Update:**

- New row added to alerts TableView
- Status column shows "UNACKNOWLEDGED" in red
- Severity column shows "SIGNIFICANT" with urgent icon

3. **Audio Feedback:**

- Urgent alert tone played once (if sound enabled)

4. **System Tray** (if minimized):

- Badge count incremented
- Desktop notification shown